SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I P. HARINI (192521402) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Harini

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	1·80	1·80	1·80	1·80
Explanation	25%	1·80	1·80	1·80	1·80
Application	20%	1·44	1·44	1·44	1·44
Organization	15%	1·8	1·8	1·8	1·8
Accuracy	10%	0·72	0·72	0·72	0·72
Clarity	5%	0·36	0·36	0·36	0·36

Total Marks 32 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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COURSE : COMPUTER NETWORK

COURSE CODE : CSA0710

ASSESMENT : COTATI CLASS TEST

37/40

...TION LAYER - COMPRESSION AND ENCRYPTION :

INTRODUCTION :

The Presentation Layer is the sixth Layer of the OSI Model. It prepares data for communication between different systems. Its main functions include data compression, encryption, decryption and data format translation.

In this problem, a multimedia application transfers a 900 MB video file. The file is compressed by 40% and encryption adds 5% overhead to the compressed file size. We need to calculate the compressed size, encrypted size, and total percentage reduction.

GIVEN DATA :

Original file size - 900 MB

Compressed rate = 40%

Encryption overhead = 5% of Compressed size.

CALCULATION OF COMPRESSED FILE SIZE :

Amount of data reduced = original size × compression rate /100

= 900 × 40 / 100

= 360 MB

Compressed file size = original size - Reduced data

= 900 - 360

= 540 MB

Therefore after compression, the video file size is reduced from 900 MB to 540 MB

CALCULATION OF ENCRYPTION OVERHEAD :

Encryption overhead = Compressed size × 5 /100

= 540 × 5 / 100

= 27 MB

Encrypted file size = Compressed size + Overhead

= 540 + 27

Encrypted file size is 567 MB

OF TOTAL PERCENTAGE REDUCTION:

Total reduction = Original size - Encrypted size

= 900 - 567

= 333 MB

Percentage reduction = (Total reduction / original size) × 100

= (333 / 900) × 100

Total percentage reduction = 37 %.

IMPORTANCE OF PRESENTATION LAYER:

Compression - Reduces file size & saves network bandwidth

Encryption - Protects data from unauthorized access

data Translation - Converts data into a format understood by the receiving system

storage efficiency - Reduces the storage space required for files

Secure Communication - helps maintain confidentiality of information

CONCLUSION:

The original file size is 900 MB. after Compression, it becomes 540 MB. Encryption add 27 MB, resulting in size of 567 MB The total percentage reduction is 37 %.

7. # APPLICATION LAYER - FTP DATA TRANSFER :

INTRODUCTION :

The Application layer is the swenth and topmost layer of the osi model. it provides network services to applications, such as file transfer, email and web browsing FTP (file transfer Protocol) is used to transfer files between computers over a network.

In this problem, 3GB of data is transfered with an effetive throughput of 120 Mbps. Each request contains 3MB of data Need to calculate total no. of request & transmission time.

GIVEN DATA :

Total data size = 3GB

Effective throughput = 120 Mbps

Data per request - 3 MB

TOTAL NO. OF REQUEST

Total data size = 3 × 1000
= 3000 MB

No. of requests = Total data size / size of each request
= 3000 / 3

Total no. of requests = 1000

CALCULATION OF TRANSMISSION TIME :

1 byte = 8 bits

Total data size = 3000 × 8
= 24,000 megabits

Transmission time = Total data size / throughput
= 24,000 / 120
= 200 seconds

(3 minutes 1 20 seconds)

assuming constant throughput & excluding additional delays

FUNCTIONS OF APPLICATION LAYER :

FILE TRANSFER: FTP helps transfer files between computers.

EMAIL SERVICES: Supports sending and receiving email through appropriate protocols.

WEB SERVICES: HTTP and HTTPS support communication between web browsers and servers.

RESOURSE SHARING: Allows users to access shared network resources.

USER INTERACTION: Provides network services required by applications.

CONCLUSION :

The total file size is 3000 MB, and each request contains 3MB. Therefore 1000 request are required. At a throughput of 120 mbps. the transmission time is 200 seconds.

DELAY IN THE OSI MODEL:

INTRODUCTION:

End to End delay is the total time taken for a packet to travel from the sender to the reciver. It includes processing, transmission and propagation delays.

GIVEN DATA:

Application layer = 4 ms

Transport layer = 3 ms

Network Layer = 5 ms

Data link Layer = 2 ms

Physical Layer = 1 ms

Propagation Layer = 18 ms

Transmission Layer = 12 ms

Calculation of Processing delay:

Total processing delay = 4 + 3 + 5 + 2 + 1

Total processing delay = 15 ms

Calculation of total End to End

Total delay = Processing delay + Propagation delay + Transmission delay

= 15 + 18 + 12

Total end to end delay = 45 ms

FUNCTIONS OF OSI LAYER:

Application Layer: Provides network services to applications.

Transport Layer: Supports end to end data delivery.

Network Layer: Routes packets to their destination

Data link Layer: Organizes data into frames and detects error.

Physical Layer: Transmits bits through cables or wireless signals.

Propagation delay: is the time taken by a signal to travel to the receiver.

delay is the time required
send all packet bits onto the
network.

CONCLUSION:

The total processing delay is 15ms. Adding the propagation delay of 18 ms and transmission delay of 12 ms gives a total end to end delay of 45 ms. It helps evaluate network performance & communication speed.

9. FRAME CALCULATION AND PROTOCOL OVERHEAD:

INTRODUCTION:

Protocol overhead is the additional information added to data during transmission. it helps with addressing, error detection, and communication between devices. Transmission efficiency indicates how much of the transmitted data is useful information.

Given Data:

useful data = 80 MB

Frame payload size - 1600 Bytes

Protocol overhead per frame 80 bytes

Assume 1MB = 1,000,000 bytes

Calculation of total no: of frames:

useful data = 80 × 1000,000

= 80,000,000 bytes

No. of frames = total useful data / Payload per frame

= 80,000,000 / 1600

Total no: of frames = 50,000

Calculation of total Protocol Overhead:

Total overhead = No. of frames × Overhead per frame

= 50,000 × 80

= 4,000,000 bytes

Converting into MB:

= 4,000,000 / 1,000,000

Total protocol overhead = 4 MB

Calculation of Transmission Efficiency:

Total transmitted data = useful data + Overhead

= 80 + 4

= 84 MB

Transmission efficiency = (Useful data / Total transmitted data) x 100

= (80/84) x 100

= 95.24 %

Effects of Protocol overhead:

INCREASES DATA SIZE: adds extra information to each frame.

CONSUMES BANDWIDTH: uses part of the available network capacity.

AFFECTS TRANSMISSION TIME: Additional bytes must be Transmitted.

Supports Communication : Provides addressing and error detection information.

Improves Reliability : Helps detect transmission error

CONCLUSION : A total of 50,000 frames is required to transmit 80MB of useful data. The total protocol overhead is 4 MB. The total transmitted data is 84 MB. Therefore the Transmission efficiency is 95.24 %.